

Tutorial of ADMET prediction of drugs and molecular properties calculation by MolAICal

Qifeng Bai

Email: molaical@yeah.net

Homepage: <https://molaical.github.io/baiqf.html>

School of Basic Medical Sciences

Lanzhou University

Lanzhou, Gansu 730000, P. R. China

1. Introduction

About 40% of drug candidates have failed in the past due to toxicity and unacceptable efficacy. In this case, the predictions of drug ADMET (**absorption, distribution, metabolism, excretion, and toxicity**) play an important role in the success of a drug candidate. In this tutorial, MolAICal is introduced to calculate ADMET and molecular properties of drugs by implement the interface package of FP-ADMET [1] tool and improved adme-pred-py under the permitted GNU General Public License v3.0 and AGPL-3.0 license. For more detailed MolAICal, please visit: <https://molaical.github.io>.

2. Materials

2.1. Software requirements

1) MolAICal: <https://molaical.github.io> or <https://molaical.gitlab.io>

Notice: Make sure the installation of MolAICal is correct!

2.2. Example files

1) All the necessary tutorial files are downloaded from:

<https://github.com/MolAICal/tutorials/tree/master/018-admet>

Part I. ADMET calculation of drugs

1. Install FP-ADMET model

Currently, only **Linux version** of MolAICal is supported in Part I tutorial.

1) Click and open the website: **DownloadModel**

Then go to the folder “**AImodels→ADMET→FPADMET→linux64**” in the open website, download a file named “fpadmet.tar.gz”

2) Move “fpadmet.tar.gz” to the folder “MolAICal-xxx/mtools” **in the MolAICal container**. Where “MolAICal-xxx” is your decompressed root directory of MolAICal **in the MolAICal container**. The folder “mtools” is the targeted directory. The movement procedures are as follows:

a) First of all, copy the files to the MolAICal container.

```
***** 1. Go to the container file system, it will enter the "/root" *****
```

```
#> molaical.exe -eset shell in
```

```
***** 2. The first part of 'cp' is from the local machine, and the second part is in the container *****
```

```
#> mv <Your machine path>/fpadmet.tar.gz /root/soft/MolAICal-xxx/mtools
```

```
***** 3. Exit container file system *****
```

```
#> exit
```

b) Secondly, enter the container virtual environment of MolAICal:

```
#> molaical.exe -eset sys run molaical  
#> cd /root/soft/MolAICal-xxx/mtools
```

Note: 'molaical' is the container name.

3) Decompress file

```
#> tar -xzf fpadmet.tar.gz
```

And go to folder named fpadmet:

```
#> cd fpadmet
```

4) install fpadmet model

```
#> chmod +x install.sh
```

```
#> ./install.sh
```

Until now, fpadmet model is installed completely.

- ✧ **Remember to use the "exit" command to quit the MolAICal virtual environment and return to the local computer for computation (primarily to avoid file-copying steps; execution within the container is also possible but requires copying data from the local computer to the MolAICal container).**

```
#> exit
```

2. Calculate ADMET of drugs

2.1. Input file format for ADMET calculation

Users should prepare the right format for the input file. One molecule occupies one line of the input file. For one line, the first column contains molecular SMILES string, the second column includes the molecular name. And the molecular name should have one space with molecular SMILES string at least (see Figure 1).

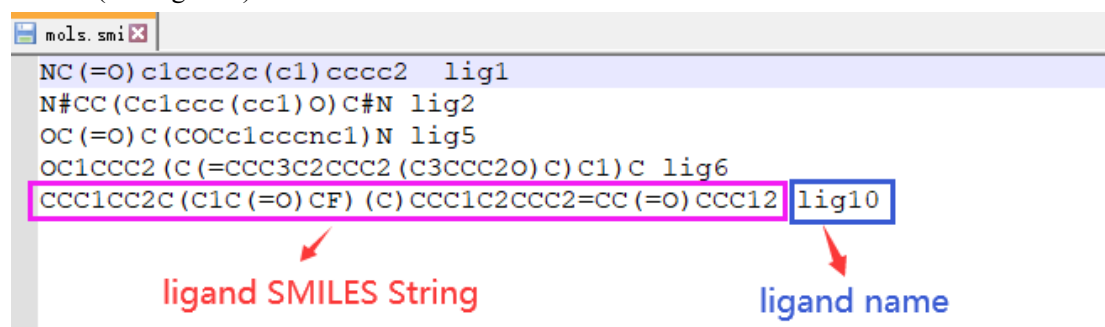


Figure 1. Content of input file

2.2. Calculate all items of ADMET from a file containing molecular SMILES

Go to 018-admet, and use the below command:

```
#> molaical.exe -model fpadmet -s all -i mols.smi -o results.dat
```

Interpretations of results: It will generate a file named “**results.dat**” that contains all items of ADMET of drug. Open file “**results.dat**”, it will show 58 item results for one drug. For example, predicting Blood–brain-barrier penetration (item 2) and pKa dissociation constant (item 50) are shown below :

```
#####
```

2: Blood–brain-barrier penetration

Predicted Confidence Credibility

lig1 Yes 0.93 0.32

```
.....
```

50: pKa dissociation constant

Predicted quantile_0.025 quantile_0.975

lig1 5.19 -1.60 13.06

```
#####
```

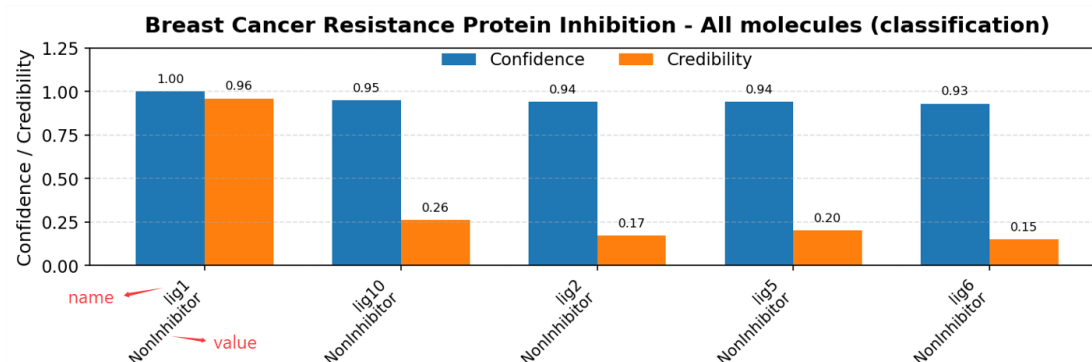
“lig1” is a drug or ligand name. The label “Yes” of compound “lig1” indicates that this compound is predicted to be suitable for blood brain barrier penetration. A confidence value of 0.93 indicates that the classifier is quite certain, and the prediction is possible to be a single label. A relatively low value of credibility (0.32) suggests that the compounds like “lig1” are not sufficiently represented in the training set, and users should deal with the prediction with caution. For the regression example of item 50, a 95% prediction interval (predictions at 0.025 and 97.5 percentiles for pKa) is computed and gives a range for the predictions on individual observation. As the original paper on FP-ADMET said: narrow prediction intervals indicate a lower uncertainty associated with the prediction [1]. For more detailed interpretations of predicted results, please check the files named “ADMETModels-doc.pdf”, “FP-ADMET.pdf” and “FP-ADMET-SI.pdf” in folder “doc” along with this tutorial document.

To visualize the results, the following command can either generate plots of 58 property values for each individual molecule (where each molecule has its own folder named after it), or generate images for each property entry across all molecules, which are saved in the folder "merge_lig":

```
#> molaical.exe -call run -c sfile -i m_admetp.py -i results.dat --merge
```

It will generate plots of 58 property values for each individual molecule in the folders that have the same names as the ligand names (e.g., lig1, lig2...). And generate images for each property entry across all molecules in the folder "merge_lig". **The results are provided as PNG images and PDF documents, with the results in the PDF documents being consistent with the image results.**

The example figure is shown below. The x-axis represents the molecular names and their corresponding ADMET values. Please refer to the calculation results when interpreting this figure:



2.3. Calculating a specified item of ADMET based on a file that contains SMILES strings

Go to [018-admet](#), and use the below command:

```
#> molaical.exe -model fpadmet -s single -i mols.smi -n 1
```

Interpretations of results: It will generate a file named “1.dat” due to selecting task item 1. If task item 2 is chosen, it will produce a file named “2.dat”. If task item 3 is chosen, it will produce a file named “3.dat”... So the prefix name of generated result file is the same as the selected task item. Open file “1.dat”, the content of “1.dat” is like below:

```
#####
Predicted Confidence Credibility
lig1 Negative 0.77 0.31
lig2 Negative 0.90 0.59
lig5 Negative 0.80 0.35
lig6 Negative 0.91 0.64
lig10 Negative 0.96 0.86
#####
```

It only shows the calculated results of “1: Anticommensal Effect on Human Gut Microbiota” item of ADMET for all molecules. For understanding all items of ADMET, please check the MolaIcal manual, Appendix in this tutorial, or the files named **“ADMETModels-doc.pdf”** in the folder doc, “FP-ADMET.pdf” and “FP-ADMET-SI.pdf” in folder “doc” along with this tutorial document.

To visualize the results, it can find the figure according to the item name in the above-generated folder “merge_lig”.

Part II. Calculations of ADME and molecular properties by MolaIcal

In this part, **Linux or Windows version** of MolaIcal is supported for this part tutorial.

In this part tutorial, it will show the usages of MolaIcal for the calculations of ADME and molecular properties.

1) If “-s” is 0, it will show all calculated molecular properties information

```
#> molaical.exe -model admepr -s 0 -i "O=C(C)Oc1ccccc1C(=O)O"
```

2) If “-s” is 1, it will show the brief general ADME information of a ligand

```
#> molaical.exe -model admepr -s 1 -i "O=C(C)Oc1ccccc1C(=O)O"
```

For example, the output is shown below:

1. Pharmacokinetics:

GI absorption: High

BBB permeant: Yes

2. Lipinski Rule of 5 Violations:

No violations found

3. Medicinal Chemistry:

PAINS filter: False

Brenk filter: True

3) If “-s” is 2, it will show basic molecular properties, for example: molecular weight, Number of Hydrogen Bond Donors, etc.

```
#> molaical.exe -model admepr -s 2 -i "O=C(C)Oc1ccccc1C(=O)O"
```

For more detailed items of molecular properties, please check MolAICal manual.

Reference

1. Venkatraman V. FP-ADMET: a compendium of fingerprint-based ADMET prediction models. J Cheminform. 2021;13(1):75.

Appendix

58 ADMET items of FP-ADMET are listed as follows:

- 1: Anticommensal Effect on Human Gut Microbiota
- 2: Blood–brain–barrier penetration
- 3: Oral Bioavailability
- 4: AMES Mutagenecity
- 5: Metabolic Stability
- 6: Rat Acute LD50
- 7: Drug-Induced Liver Inhibition
- 8: HERG Cardiotoxicity
- 9: Haemolytic Toxicity
- 10: Myelotoxicity
- 11: Urinary Toxicity
- 12: Human Intestinal Absorption
- 13: Hepatic Steatosis
- 14: Breast Cancer Resistance Protein Inhibition
- 15: Drug-Induced Cholestasis
- 16: Human multidrug and toxin extrusion Inhibition
- 17: Toxic Myopathy
- 18: Phospholipidosis
- 19: Human Bile Salt Export Pump Inhibition
- 20: Organic anion transporting polypeptide 1B1 binding
- 21: Organic anion transporting polypeptide 1B3 binding
- 22: Organic anion transporting polypeptide 2B1 binding
- 23: Phototoxicity human
- 24: Phototoxicity in vitro
- 25: Respiratory Toxicity
- 26: P-glycoprotein Inhibition
- 27: P-glycoprotein Substrate
- 28: Mitochondrial Toxicity
- 29: Carcinogenicity
- 30: DMSO Solubility
- 31: Human Liver Microsomal Stability
- 32: Human Plasma Protein Binding
- 33: hERG Liability
- 34: Organic Cation Transporter 2 Inhibition
- 35: Drug-induced Ototoxicity
- 36: Rhabdomyolysis
- 37: T_{1/2} Human
- 38: T_{1/2} Mouse
- 39: T_{1/2} Rat

- 40: Cytotoxicity HepG2 cell line
- 41: Cytotoxicity NIH 3T3 cell line
- 42: Cytotoxicity HEK 293 cell line
- 43: Cytotoxicity CRL-7250 cell line
- 44: Cytotoxicity HaCat cell line
- 45: CYP450 1A2 Inhibition
- 46: CYP450 2C19 Inhibition
- 47: CYP450 2C9 Inhibition
- 48: CYP450 2D6 Inhibition
- 49: CYP450 3A4 Inhibition
- 50: pKa dissociation constant
- 51: logD Distribution coefficient (pH 7.4)
- 52: logS
- 53: Drug affinity to human serum albumin
- 54: MDCK permeability
- 55: 50% hemolytic dose
- 56: Skin penetration
- 57: CYP450 2C8 Inhibition
- 58: Aqueous Solubility (in phosphate saline buffer)